

Programme : Diploma in Cloud Computing and Big Data	
Course Code : 6272A	Course Title: Fundamentals of Cloud Computing
Semester : 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To impart basic introduction to cloud computing and cloud architecture
- To provide an understand on virtualization and cloud database
- To impart awareness on various industrial cloud platforms

Course Prerequisites:

Topic	Course Code	Course name	Semester
Computer Fundamentals	1008	Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Outline fundamental concepts of Cloud Computing.	14	Understanding
CO2	Explain Cloud Architecture and models	15	Understanding
CO3	Summarize the basic concepts of Virtualization and Cloud database	16	Understanding
CO4	Describe different public cloud platforms	13	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						2
CO2	3						2
CO3	3						2
CO4	3						2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline fundamental concepts of Cloud Computing		
M1.01	Summarize the history of Cloud Computing	4	Understanding
M1.02	Outline the basic Terminology in cloud computing	3	Understanding
M1.03	Compare Parallel and Distributed Computing	3	Understanding
M1.04	Describe Elements of Distributed Computing	4	Understanding
Contents: History of Cloud Computing- Salesforce.com,Amazon, Google Apps Basic Concepts and Terminology: Defining a cloud, Goals and Benefits, Risks and Challenges, Roles and Boundaries, Cloud Characteristics, Types of cloud computing- Virtualization,Service Oriented Architecture (SOA),Grid Computing ,Utility Computing Principles of Parallel and Distributed Computing: Parallel vs. Distributed computing, Elements of Parallel Computing- Hardware architectures for parallel processing, approaches to parallel programming, Levels of parallelism, Elements of Distributed Computing- General concepts and definitions, Components of a distributed system			
CO2	Explain Cloud Architecture and models		

M2.01	Explain Cloud Architecture	3	Understanding
M2.02	Classify Cloud delivery models	5	Understanding
M2.03	Compare Cloud delivery models	3	Understanding
M2.04	Classify Cloud deployment models	4	Understanding
	Series Test – I	1	

Contents:

Cloud Computing Architecture

Cloud Delivery Models- The cloud reference model - Infrastructure-as-a-service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), Comparing Cloud Delivery Models, Combining Cloud Delivery Models

Cloud Deployment Models- Public Clouds, Community Clouds, Private Clouds, Hybrid Clouds.

CO3	Summarize the basic concepts of virtualization and cloud database		
M3.01	Explain the principle of Virtualization in cloud platform	4	Understanding
M3.02	Summarize the characteristics and advantages of virtualization	3	Understanding
M3.03	Explain the basic concepts of cloud database	4	Understanding
M3.04	Describe cloud database architecture and CAP theorem	5	Understanding
	Series Test – II	1	

Virtualization: Introduction, Role of virtualization, Hypervisor, Virtualization versus Traditional Approach, Characteristics of virtualized environments, Advantages of virtualization.

Cloud Database -Introduction, Non-Relational Data Models, Transactions in Cloud Databases, Advantages of Cloud Database, Cloud data base architecture, CAP Theorem

CO4	Describe different public cloud platforms		
M4.01	Outline Specialized Cloud Mechanisms	4	Understanding

M4.02	Explain Amazon web services,	3	Understanding
M4.03	Describe Google App Engine	3	Understanding
M4.04	Describe Microsoft Azure	3	Understanding
	Series Test – II	1	

Contents:

Specialized Cloud Mechanisms: Automated Scaling Listener, Load Balancer, SLA Monitor, Pay-Per-Use Monitor, Failover System, Resource Cluster, Multi-Device Broker, State Management Database.

Cloud Platforms in Industry: Amazon web services - Compute services - Storage services - Communication services.

Google App Engine – Architecture.

Microsoft Azure: Azure core concept, SQL Azure.

Text / Reference

T/R	Book Title/Author
T1	Erl, Thomas, Ricardo Puttini, and Zaigham Mahmood. Cloud computing: concepts, technology, & architecture. Pearson Education, 2013.
T2	Rajkumar Buyya, Christian Vecchiola and Thamari Selvi S, —Mastering in Cloud Computing, McGraw Hill Education (India) Private Limited, 2013.
T3	Fundamentals of Cloud Computing ,Prasant Kumar Pattnaik , Manas Ranjan Kabat , Souvik Pal. VIKAS PUBLISHING HOUSE PVT. LTD. 2014
R1	Rishabh Sharma, Cloud Computing Fundamentals, Industry Approaches and Trends, Wiley-2015

Online Resources

Sl.No	Website Link
1	https://docs.aws.amazon.com/index.html?nc2=h_ql_doc#lang%2Fen_us
2	https://cloud.google.com/docs/
3	https://azure.microsoft.com/en-in/